

Noise Addition

Noise Type

- Random Noise
- Acquisition Footprint
- Migration Artifacts
- Residual Statics

Gaussian Noise

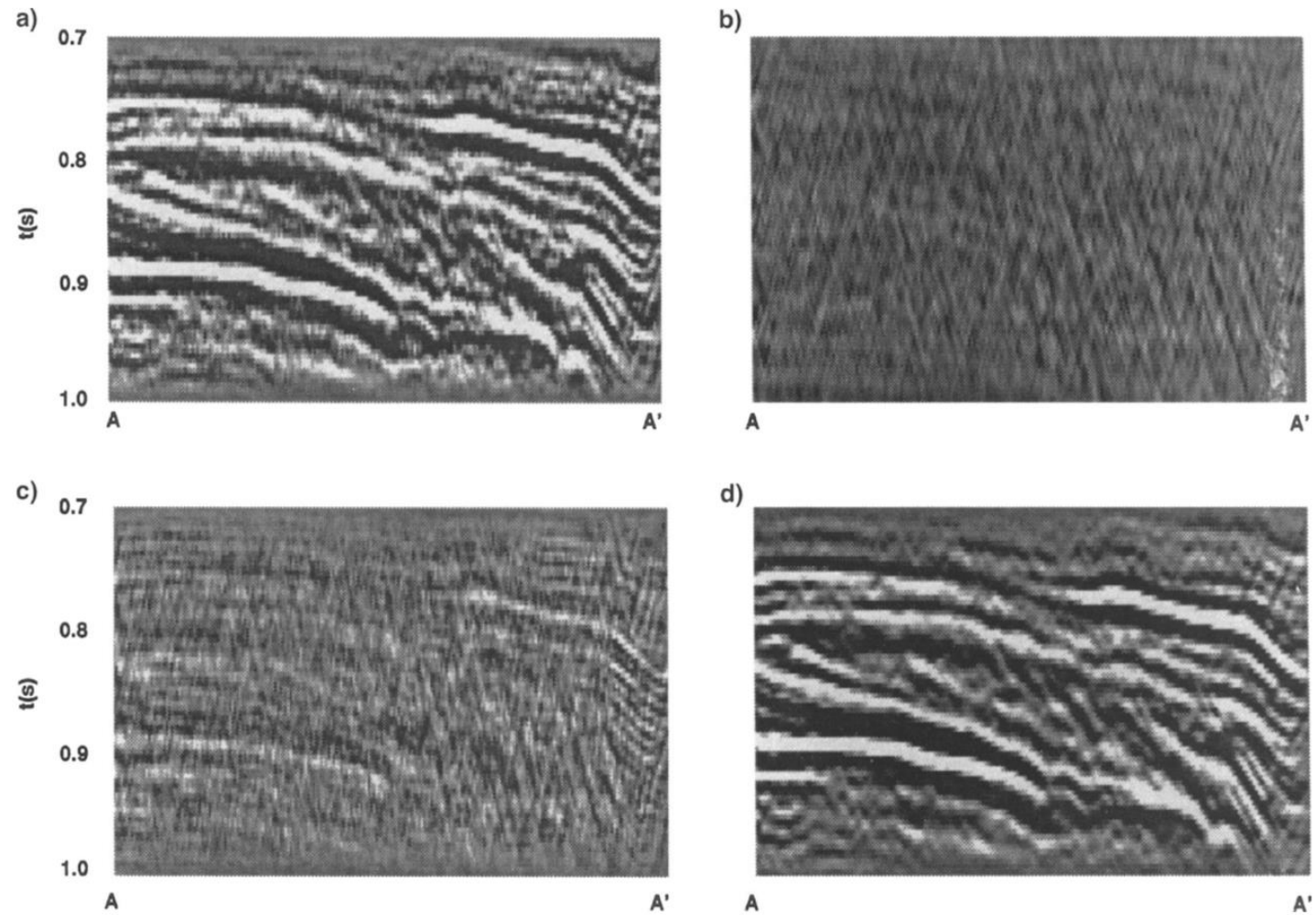
```
4 def generate_filtered_gaussian_noise(shape, dtype, sigma_filter, seed):
5     """Generates the base filtered noise (Java-style replication)."""
6     print(f"Generating base filtered Gaussian noise (sigma={sigma_filter:.4f}, seed={seed})...")
7     rng_state = np.random.get_state() # Use numpy's RNG state
8     np.random.seed(seed)
9     g_noise = 2.0 * (np.random.rand(*shape) - 0.5)
10    np.random.set_state(rng_state) # Restore numpy's RNG state
11    g_noise = g_noise.astype(dtype)
12
13    print(f"Applying Gaussian filter (sigma={sigma_filter:.4f}) with 1st derivative along axis 2...")
14    try:
15        input_dtype = g_noise.dtype
16        if np.issubdtype(input_dtype, np.integer):
17            g_noise_float = g_noise.astype(np.float64)
18        else:
19            g_noise_float = g_noise
20        filtered_noise = gaussian_filter(g_noise_float, sigma=sigma_filter, order=(0, 0, 1), mode='nearest')
21        filtered_noise = filtered_noise.astype(dtype)
22    except Exception as e:
23        print(f"Error during Gaussian filtering: {e}")
24        raise
25    return filtered_noise
```

Open

Acquisition Footprint

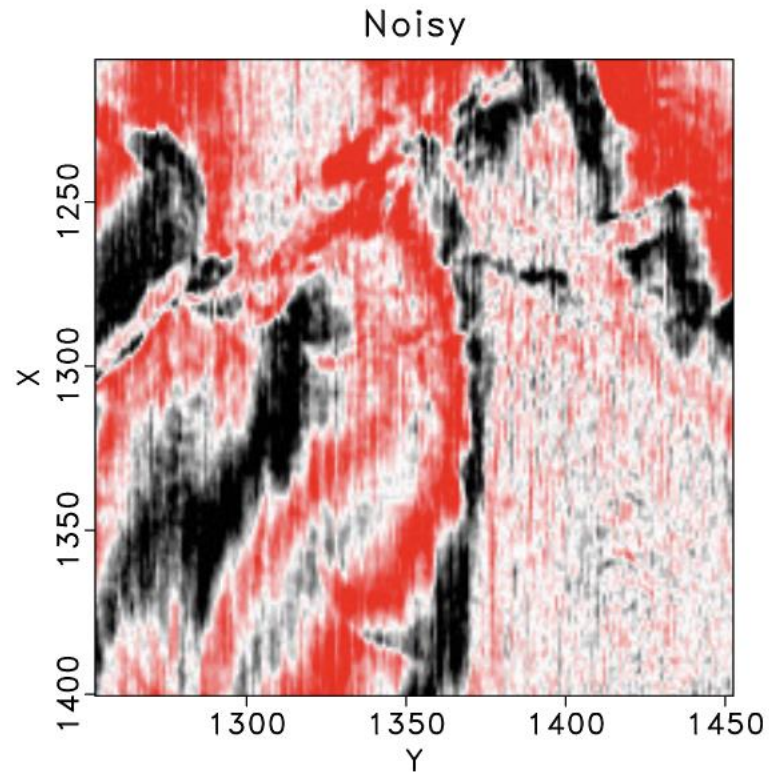
- We define the acquisition footprint to be any pattern of noise that is highly correlated to the geometric distribution of sources and receivers on the earth's surface. (Marfurt et al., 1998)

Acquisition Footprint

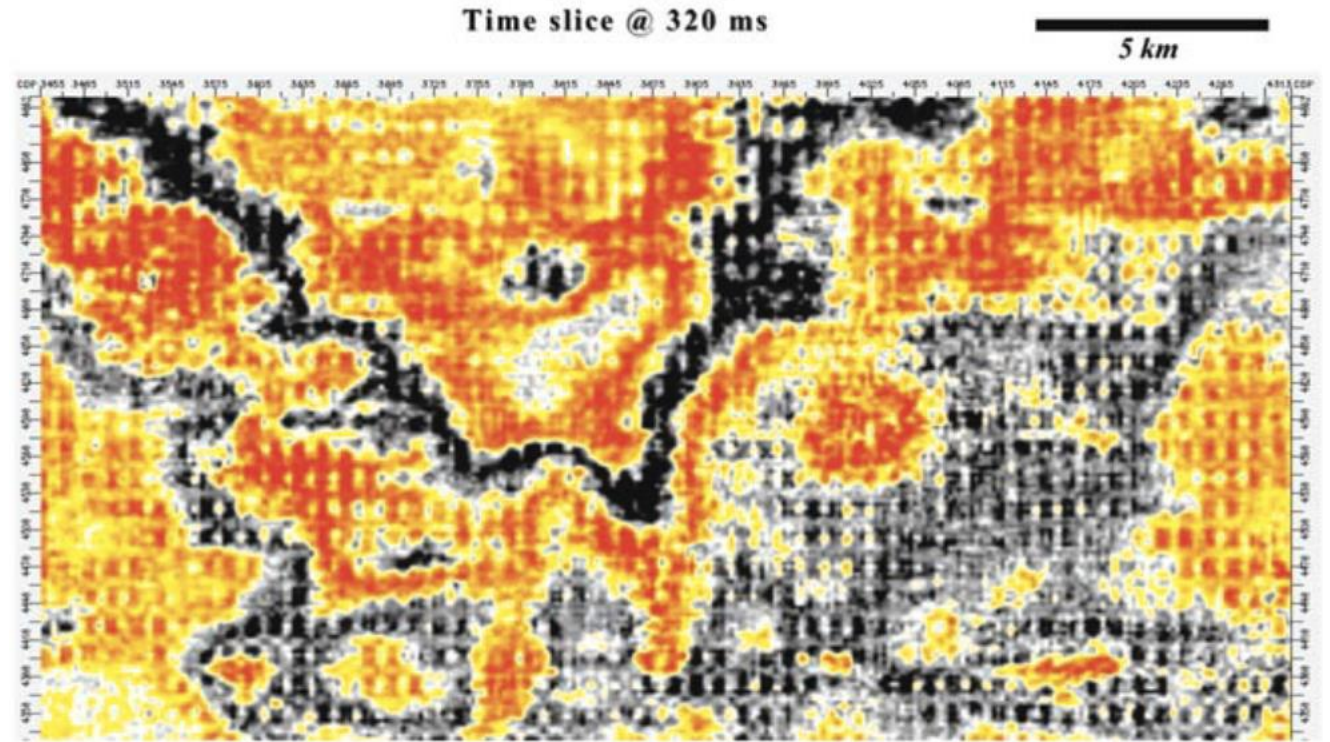


(Fig 1: Marfurt et al., 1998)

Acquisition Footprint

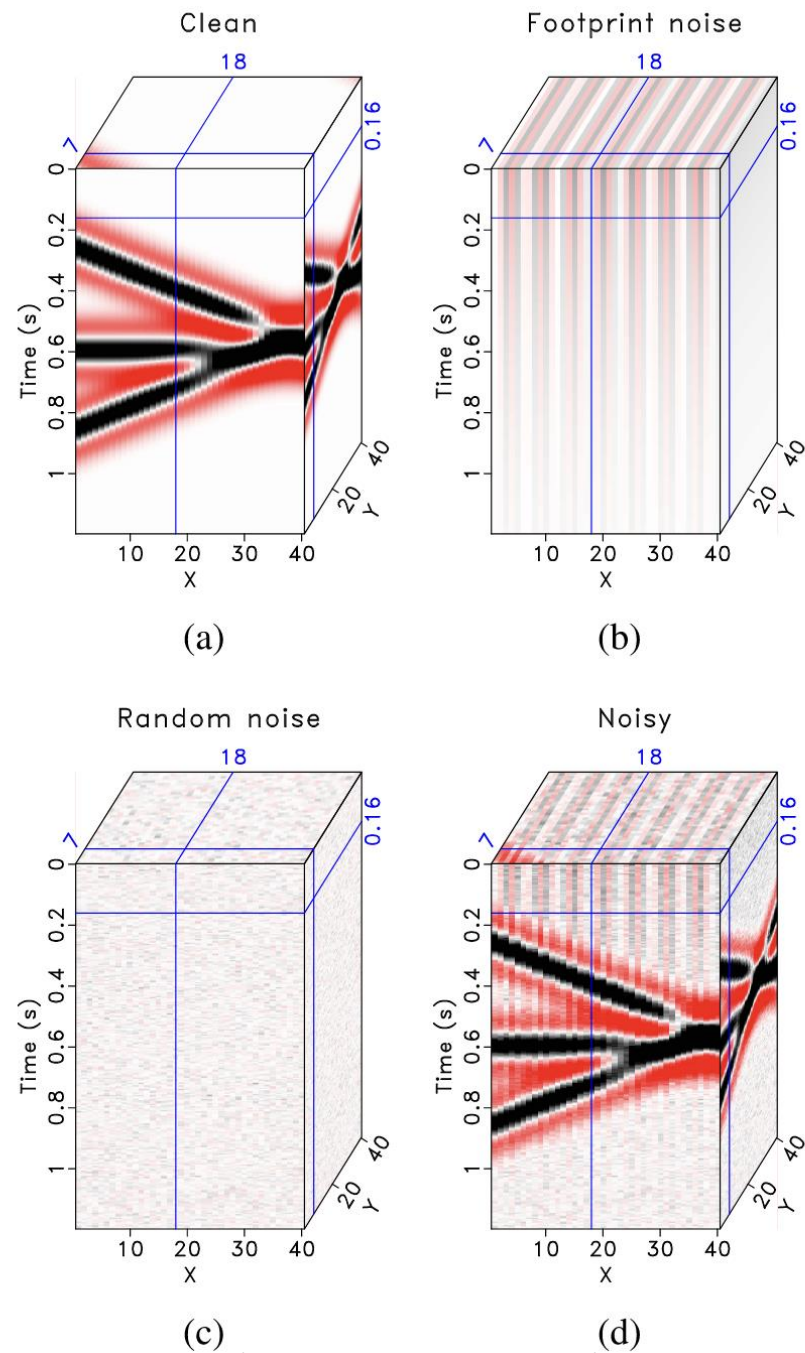


(Fig 2: Chen et al., 2022)



(Fig 3: Al-Bannagi et al., 2005)

Acquisition Footprint

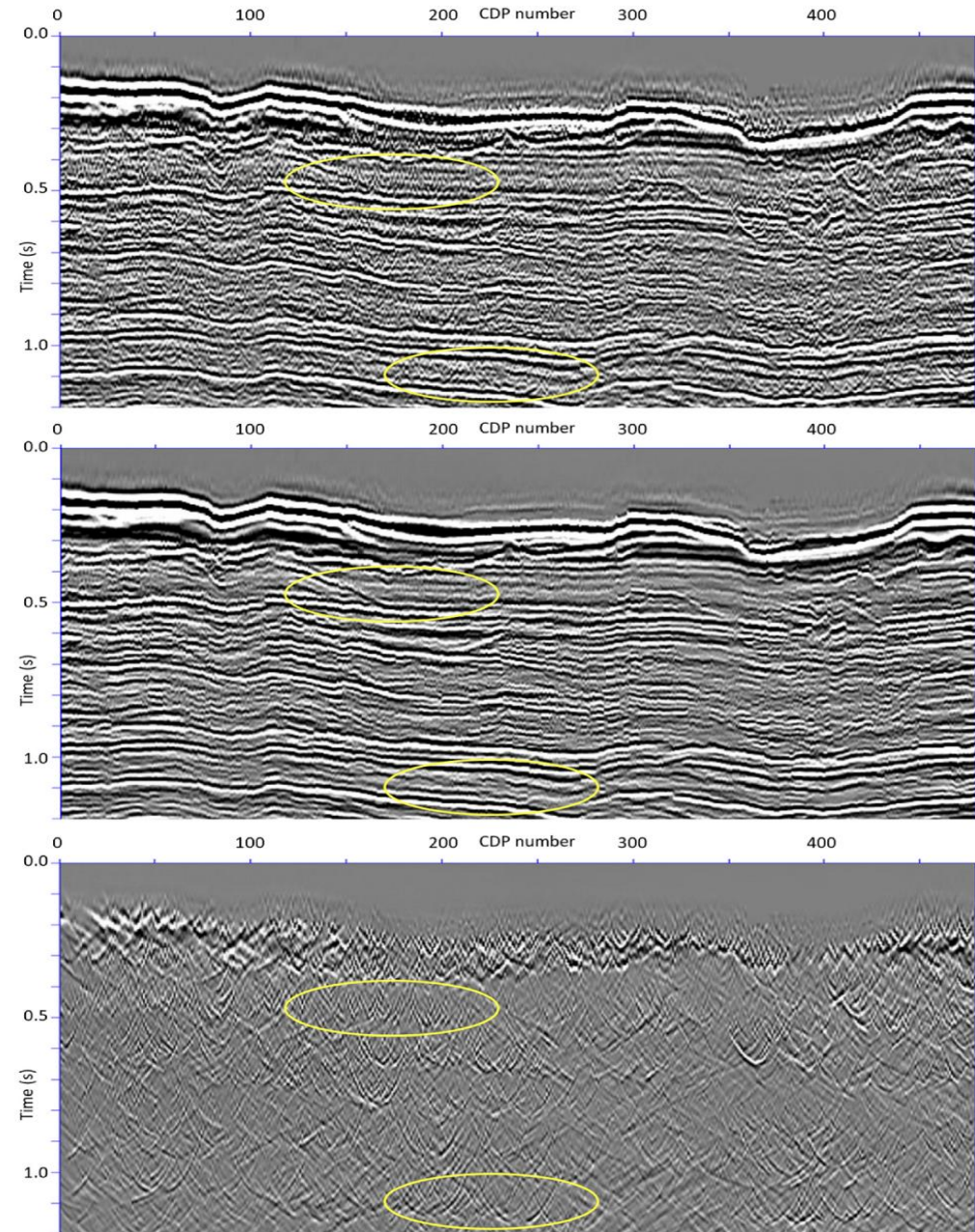


(Fig 4: Chen et al., 2022)

Migration Artifacts

- Artifacts in migrated seismic data can be caused by acquisition geometry, noise or non-primary wavefields in the data, the migration algorithm itself or errors in the velocity model. (Abma et al., 1999)

Migration Artifacts



(Fig 5: Yoo & Zwartjes, 2022)

Migration Artifacts

The smile-only segment of a section, in Fig. 3b is constructed based on the geometry of the reflection-only segment within that section meaning the artifact apex is positioned at the location of a reflector. At every location along a reflector it is randomly decided whether to place a migration smile or not. For each smile event that is placed, the wavelet, dominant frequency, opening angle and amplitude are all randomly chosen. (Yoo & Zwartjes, 2022)

Migration Artifacts



(a)



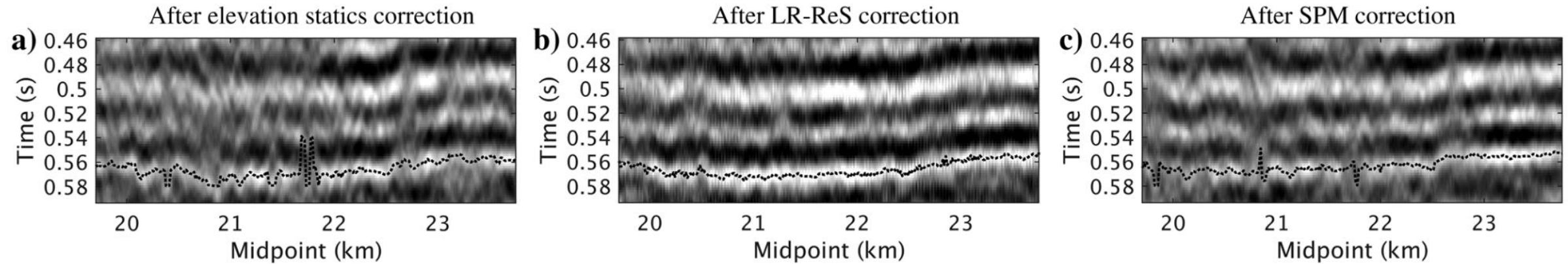
(b)

(Fig 6: Yoo & Zwartjes, 2022)

Residual Statics

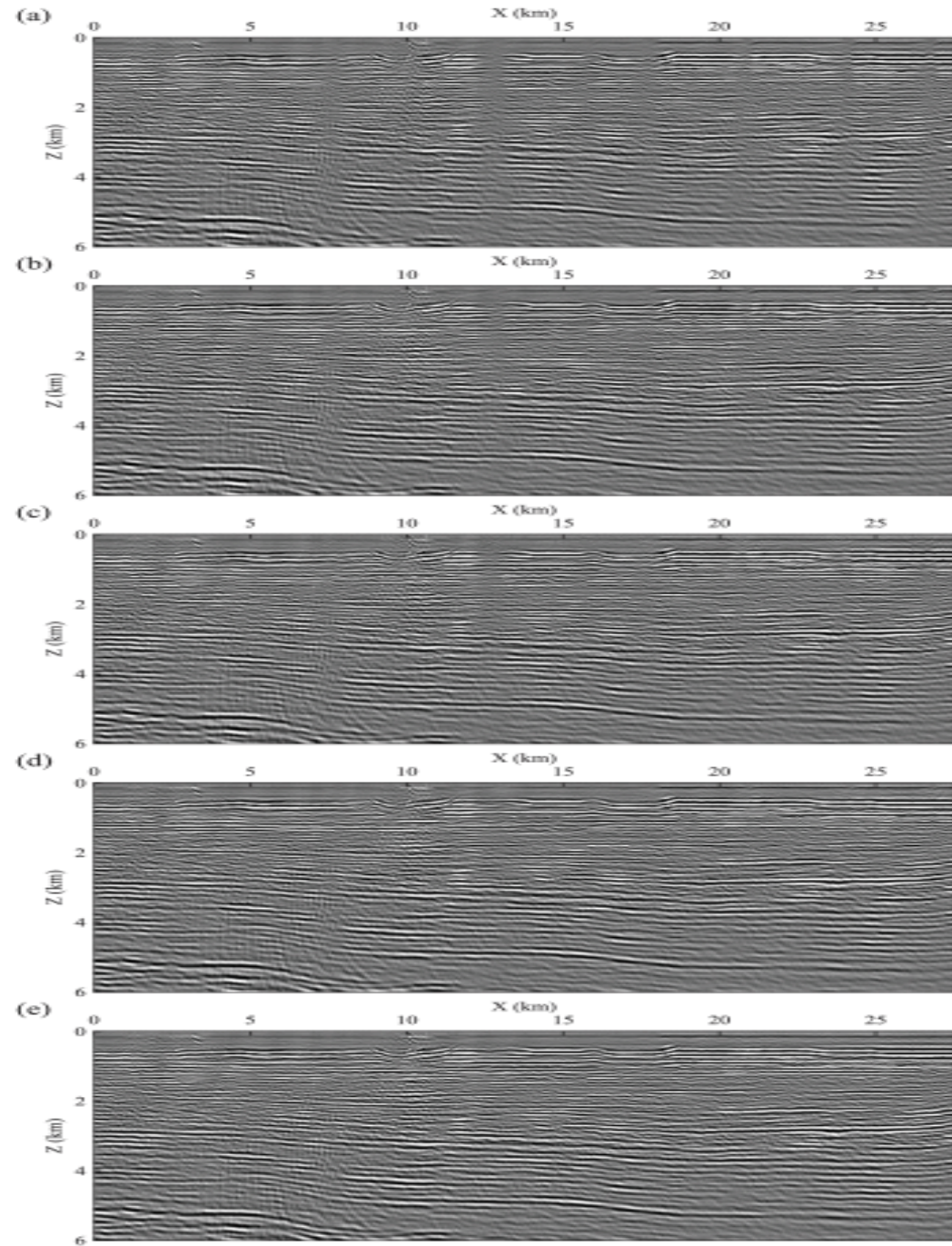
- Land seismic data sets often are challenged by near-surface weathering layers. They can lead to wave propagation effects that manifest themselves mostly as undesired time shifts, commonly called “statics”. (Taner et al., 1974)

Residual Statics



(Fig 7: Alfaraj et al., 2023)

Interpolation

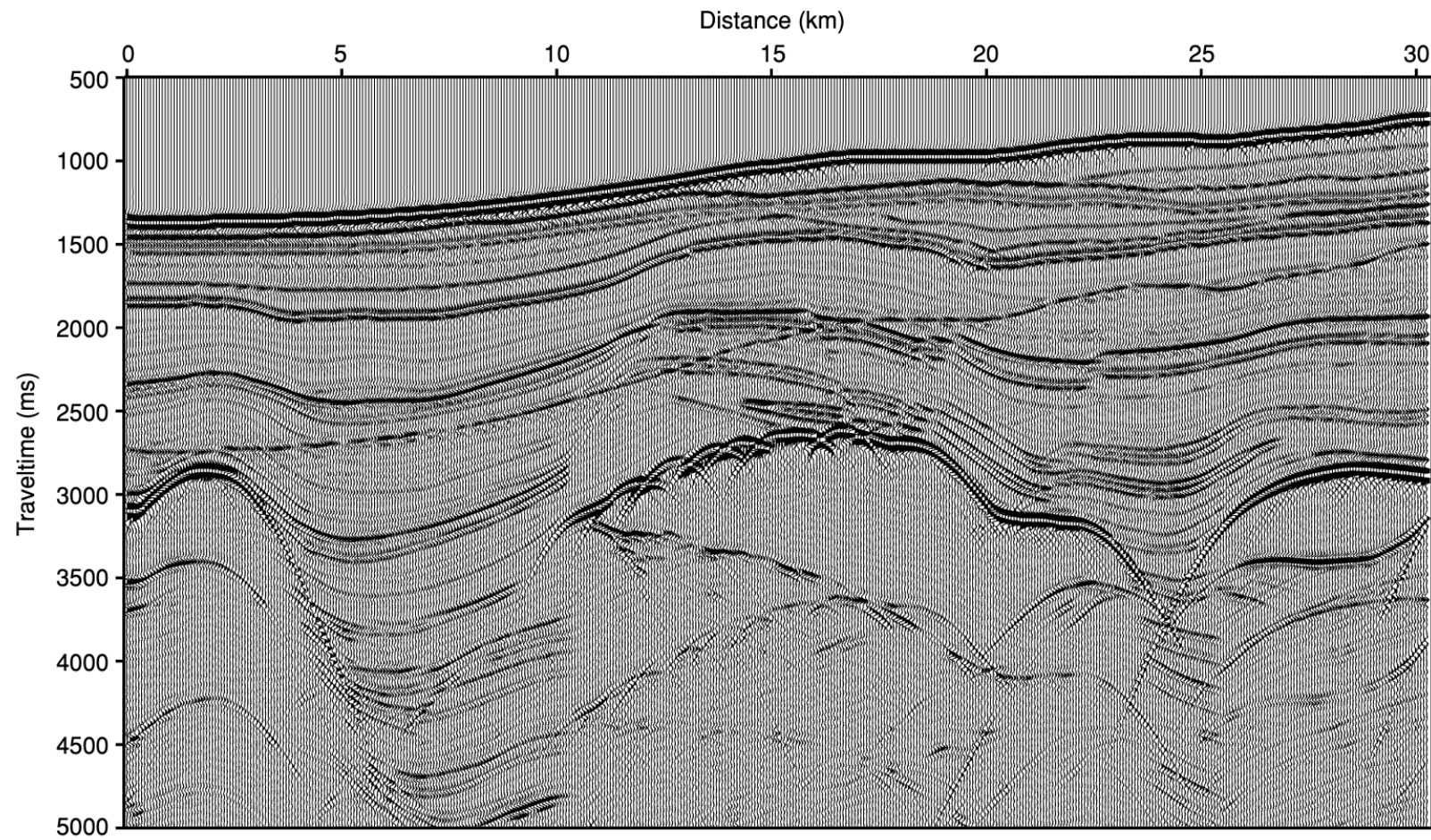


(Fig 8: Huang et al., 2022)

Multiples

- The limited size of the spatial aperture of a seismic gather causes multiple- and primary-reflection energy to spread out and cross-hatch the image of the parabolic Radon transform. (Wang, 2003)

Multiples



(Fig 9: Wang, 2003)

Reference

- Abma, R., Sun, J., & Bernitsas, N. (1999). Antialiasing methods in Kirchhoff migration. *GEOPHYSICS*, 64(6), 1783–1792. <https://doi.org/10.1190/1.1444684>
- Alfaraj, A. M., Verschuur, D. J. (Eric), & Herrmann, F. J. (2023). Low-rank-based residual statics estimation and correction. *GEOPHYSICS*, 88(3), V215–V231. <https://doi.org/10.1190/geo2022-0415.1>
- Chen, W., Saad, O. M., Wang, H., & Chen, Y. (2022). Statistics-Guided Residual Dictionary Learning for Footprint Noise Removal. *IEEE Transactions on Geoscience and Remote Sensing*, 60, 1–11. <https://doi.org/10.1109/TGRS.2021.3070903>
- Huang, H., Wang, T., Cheng, J., Xiong, Y., Wang, C., & Geng, J. (2022). Self-Supervised Deep Learning to Reconstruct Seismic Data With Consecutively Missing Traces. *IEEE Transactions on Geoscience and Remote Sensing*, 60, 1–14. <https://doi.org/10.1109/TGRS.2022.3148994>
- Marfurt, K. J., Scheet, R. M., Sharp, J. A., & Harper, M. G. (1998). Suppression of the acquisition footprint for seismic sequence attribute mapping. *GEOPHYSICS*, 63(3), 1024–1035. <https://doi.org/10.1190/1.1444380>
- Taner, M. T., Koehler, F., & Alhilali, K. A. (1974). ESTIMATION AND CORRECTION OF NEAR-SURFACE TIME ANOMALIES. *GEOPHYSICS*, 39(4), 441–463. <https://doi.org/10.1190/1.1440441>
- Wang, Y. (2003). Multiple attenuation: Coping with the spatial truncation effect in the Radon transform domain. *Geophysical Prospecting*, 51(1), 75–87. <https://doi.org/10.1046/j.1365-2478.2003.00355.x>
- Yoo, J., & Zwartjes, P. (2022). Attenuation of seismic migration smile artifacts with deep learning. *Artificial Intelligence in Geosciences*, 3, 123–131. <https://doi.org/10.1016/j.aiig.2022.11.002>

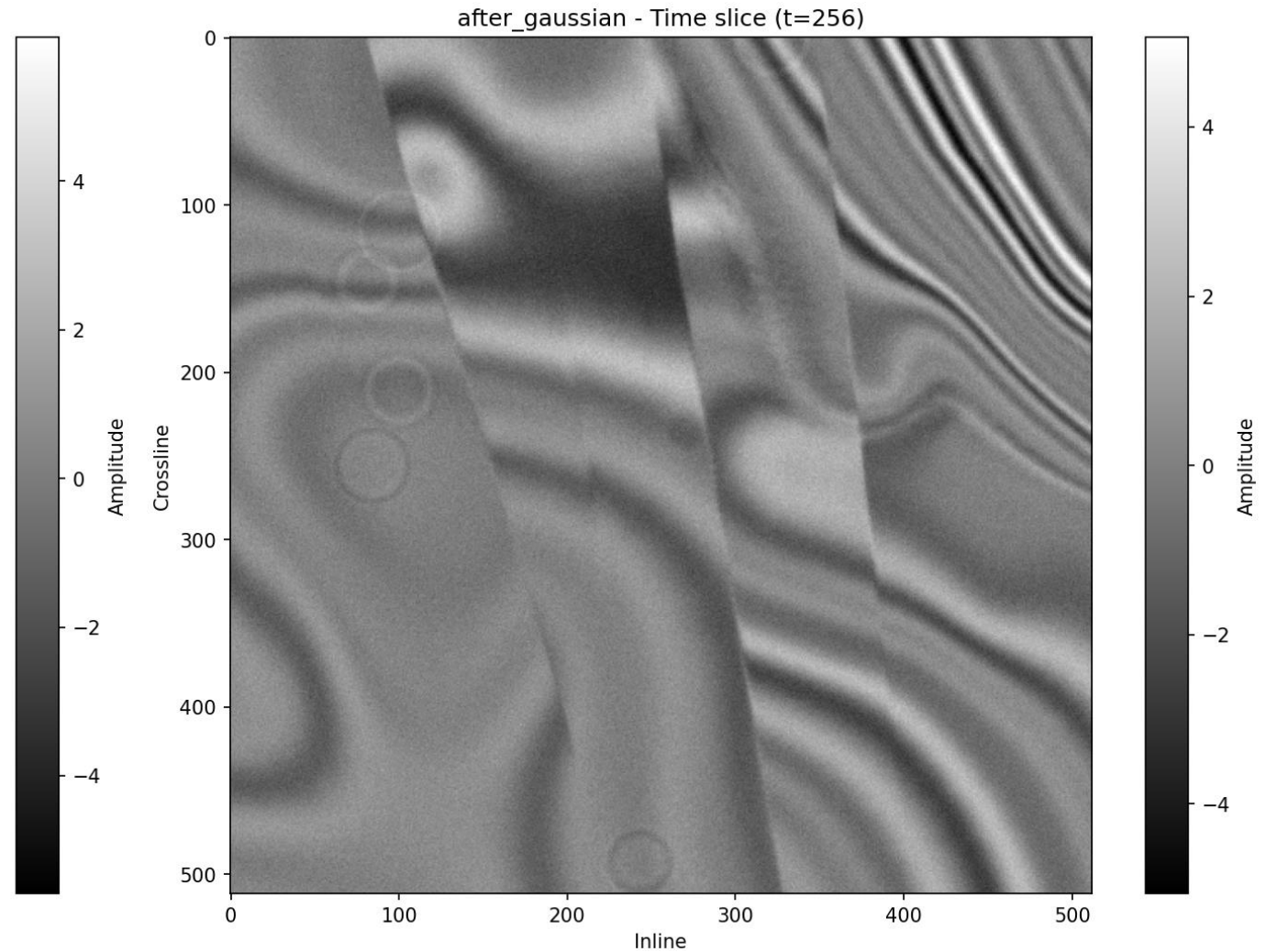
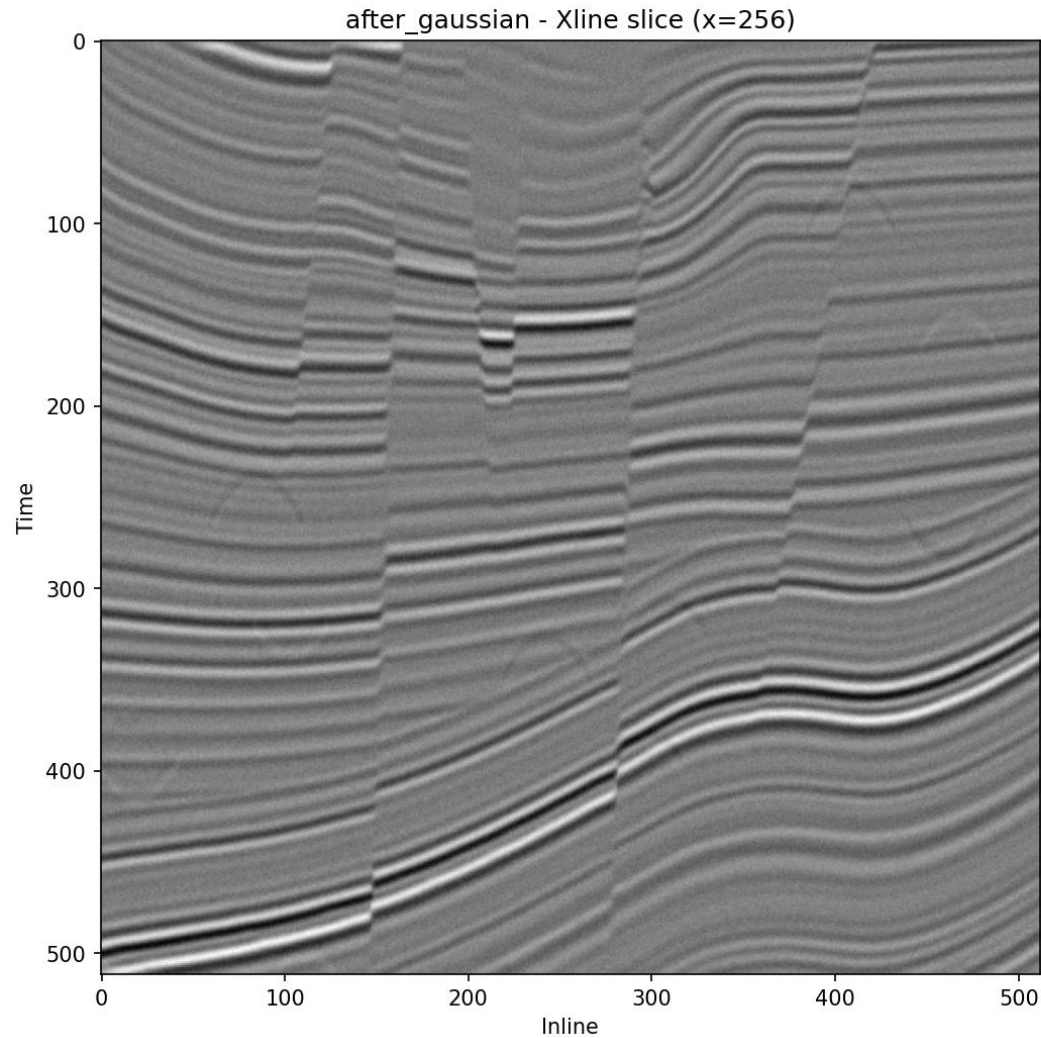
Display of noisy data

! noise_config.yaml ×

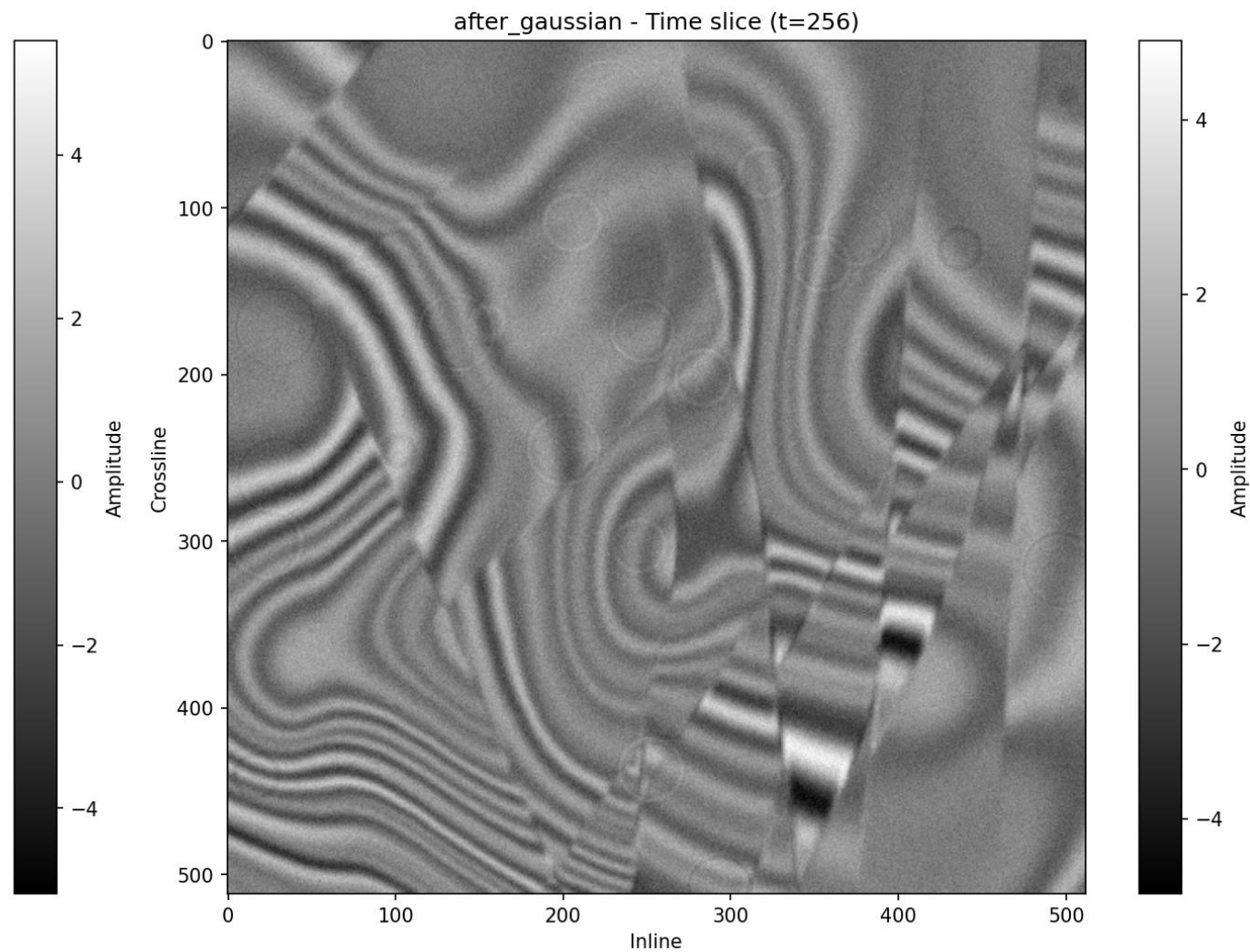
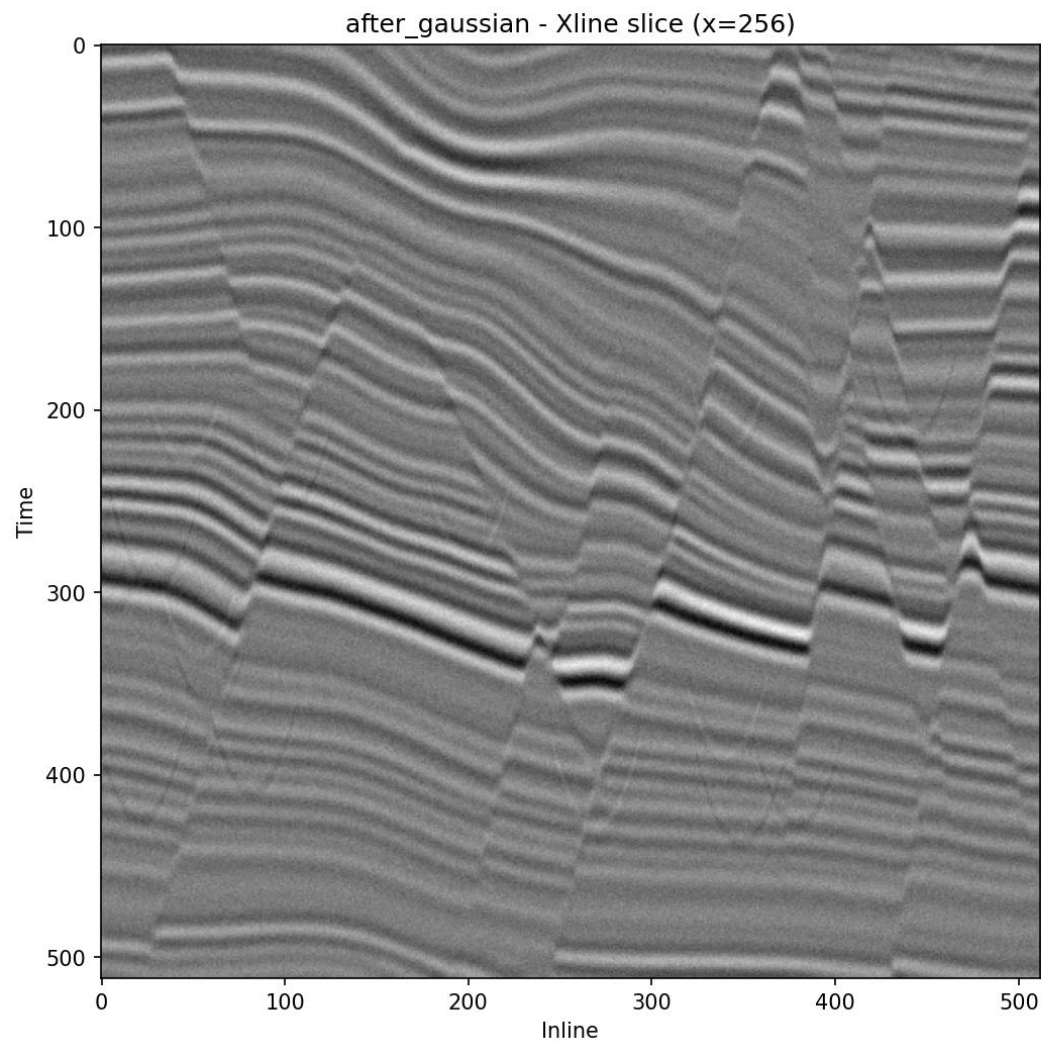
! noise_config.yaml

```
17 # 高斯噪声配置 (必加项)
18 gaussian_noise:
19   enabled: true      # 是否启用 (必须为true)
20   nrms:
21     min: 0.1         # 最小nrms值
22     max: 0.3         # 最大nrms值
23
24 # 采集足迹噪声配置
25 acquisition_footprints:
26   enabled: true      # 是否启用
27   # 固定参数 (不随机化)
28   alpha: 0.005       # 时间衰减因子
29   random_min_val: -1.0 # 随机分量最小值
30   random_max_val: 1.0 # 随机分量最大值
31   no_smoothing: false # 是否禁用高斯平滑
32
33 # 随机参数范围
34 noise_amplitude:
35   min: 1.0           # 噪声振幅最小值
36   max: 2.0           # 噪声振幅最大值
37 x_period:
38   min: 1             # Inline方向周期最小值
39   max: 2             # Inline方向周期最大值
40 y_period:
41   min: 1             # Crossline方向周期最小值
42   max: 2             # Crossline方向周期最大值
43 gaussian_sigma:
44   min: 1.0           # 高斯平滑sigma最小值
45   max: 3.0           # 高斯平滑sigma最大值
46 noise_direction: 'random' # 噪声方向: 'x', 'y', 'xy' 或 'random' (随机选择)
47
48 # 偏移伪影噪声配置
49 migration_artifacts:
50   enabled: true      # 是否启用
51   # 固定参数 (不随机化)
52   peak_prominence: 0.15 # 波峰/波谷检测阈值
53   max_artifact_radius: 25 # 最大伪影半径
54
55 # 随机参数范围
56 n_artifacts:
57   min: 100           # 伪影数量最小值
58   max: 200           # 伪影数量最大值
59 curvature:
60   min: 0.005         # 曲率最小值
61   max: 0.1           # 曲率最大值
62 artifact_direction: 'random' # 伪影方向: 'up', 'down', 'both' 或 'random' (随机选择)
```

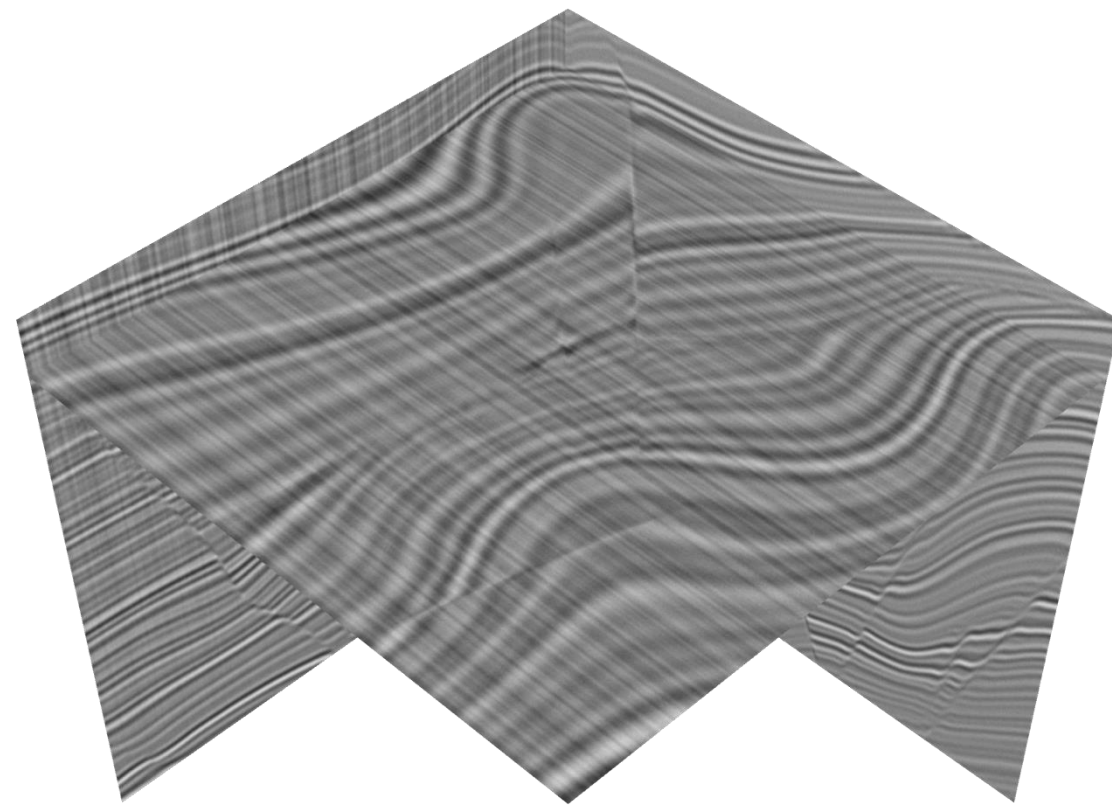
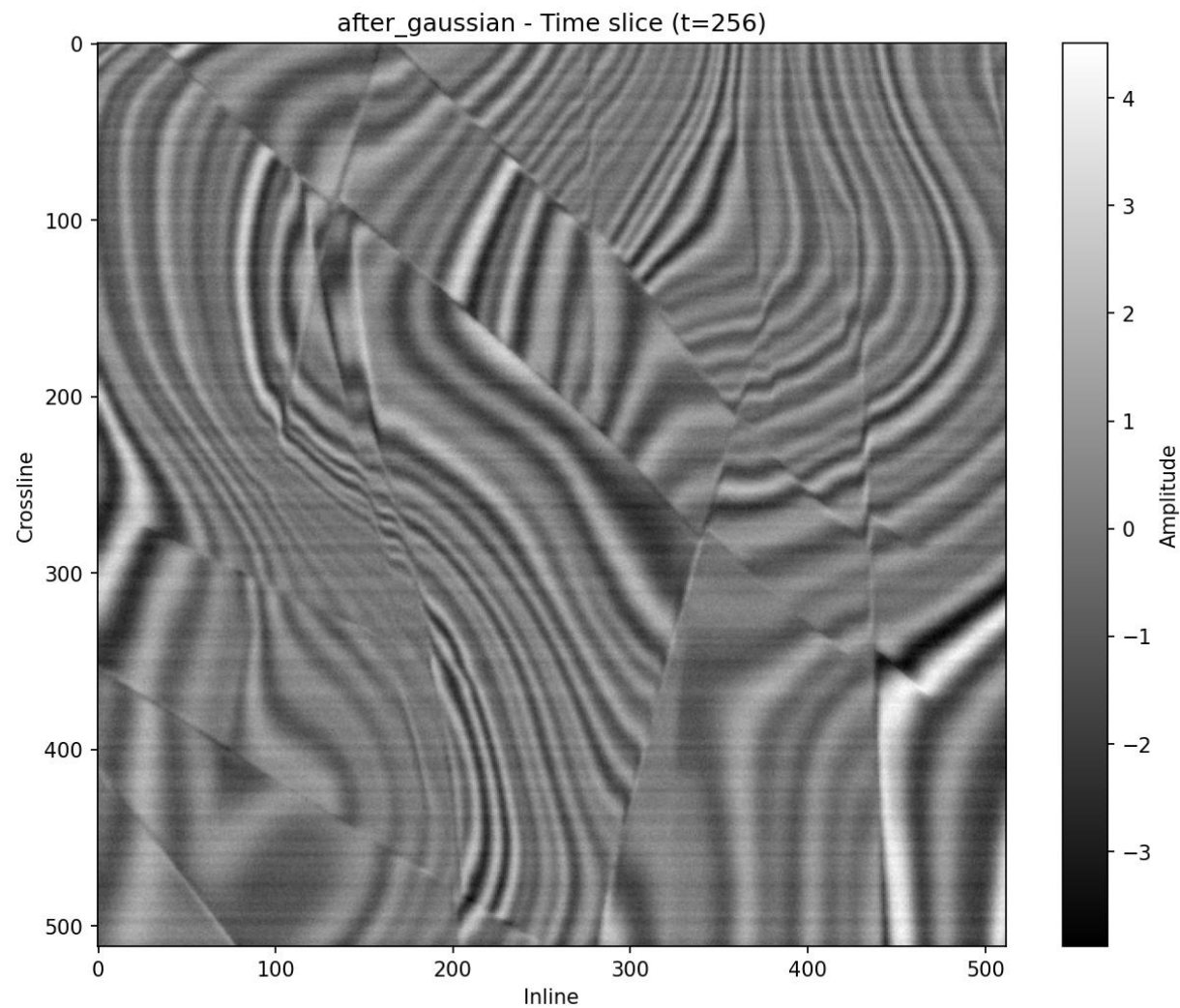
Artifacts + Gaussian Noise



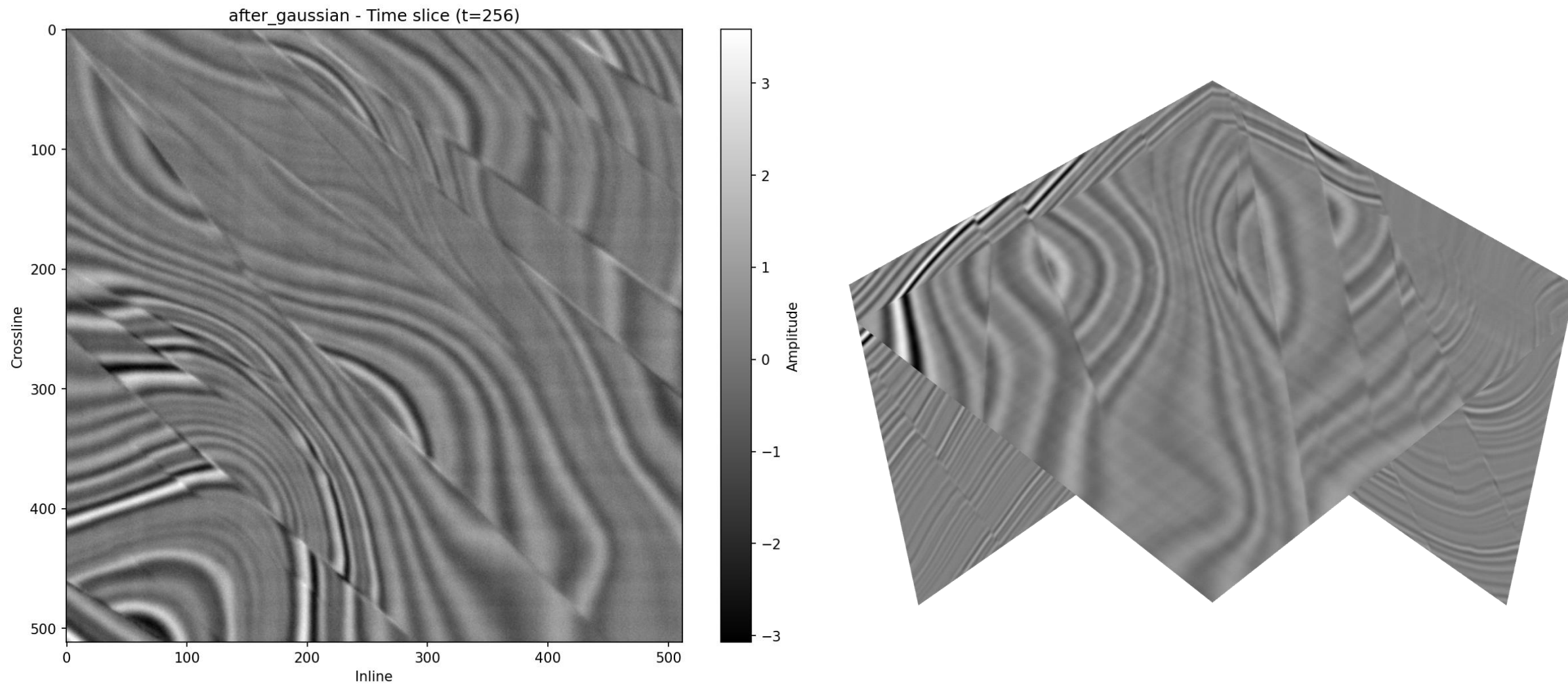
Artifacts + Gaussian Noise



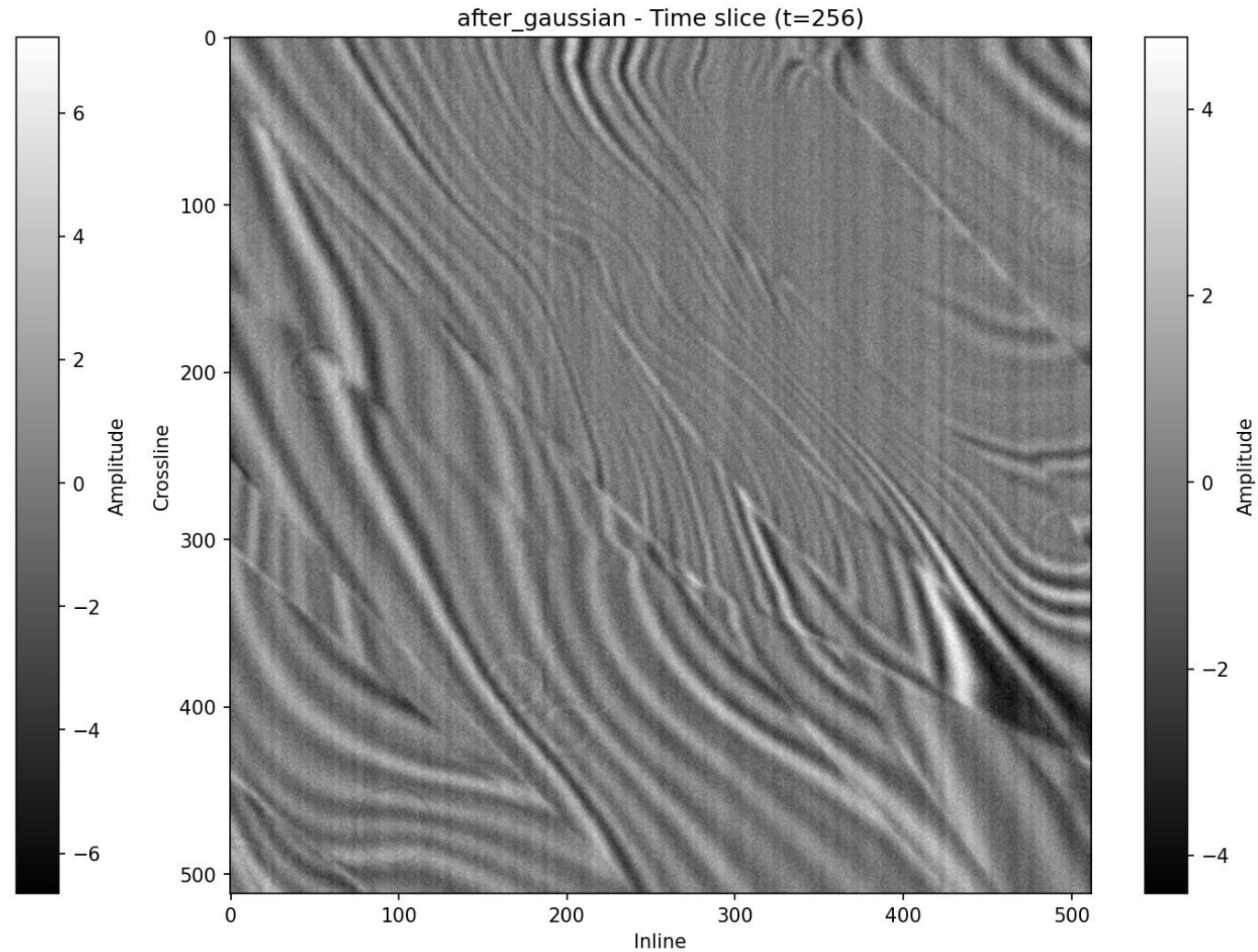
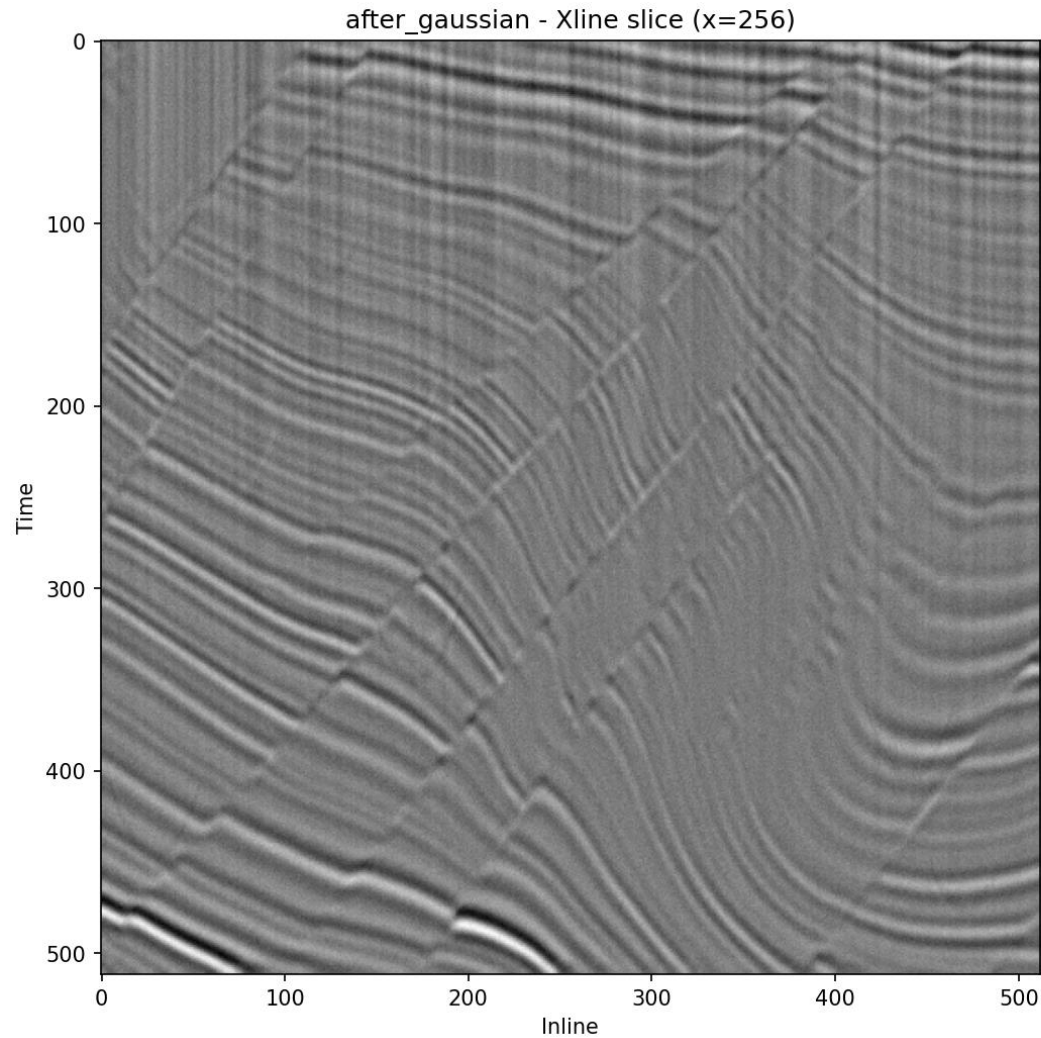
Footprints + Gaussian Noise



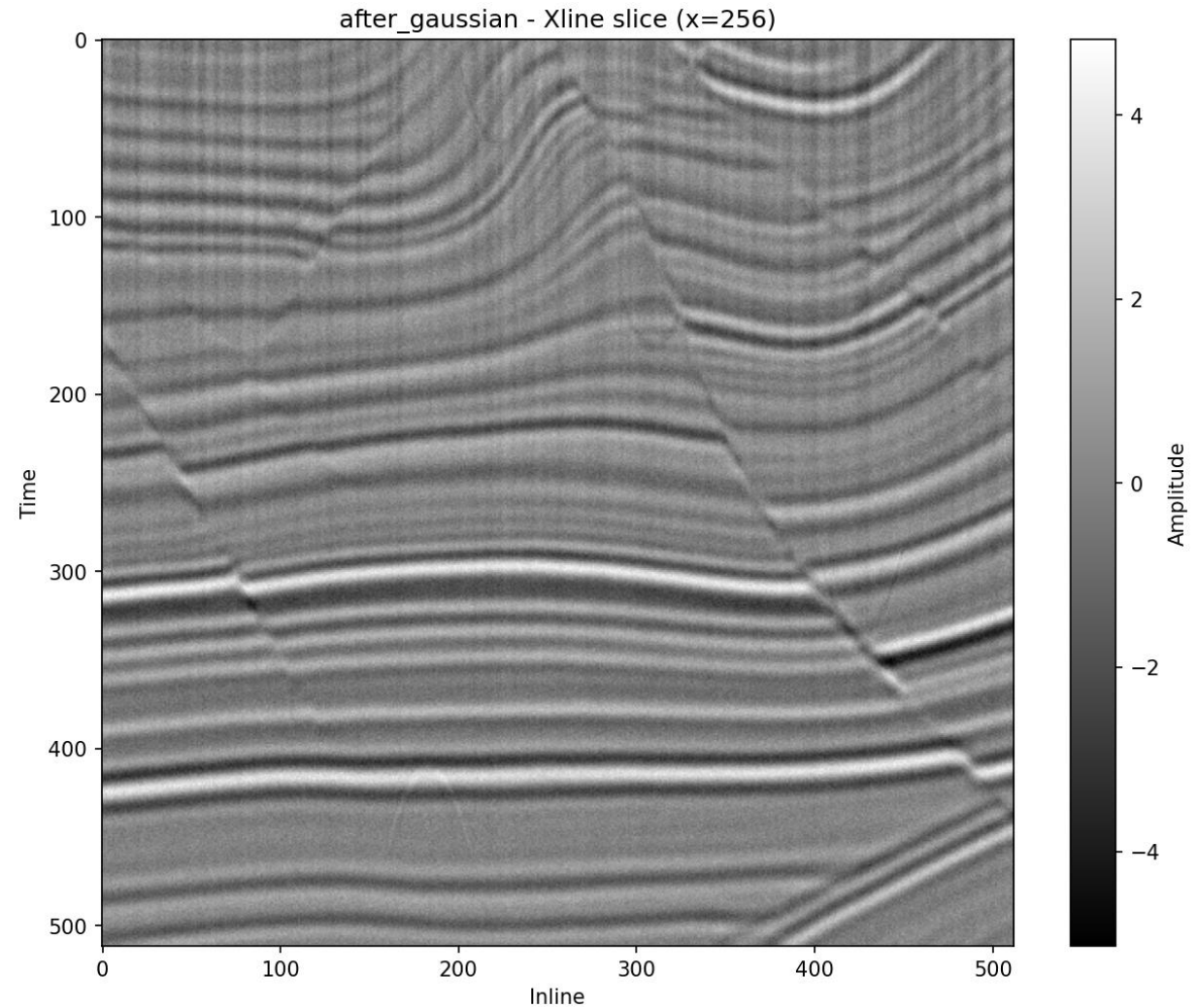
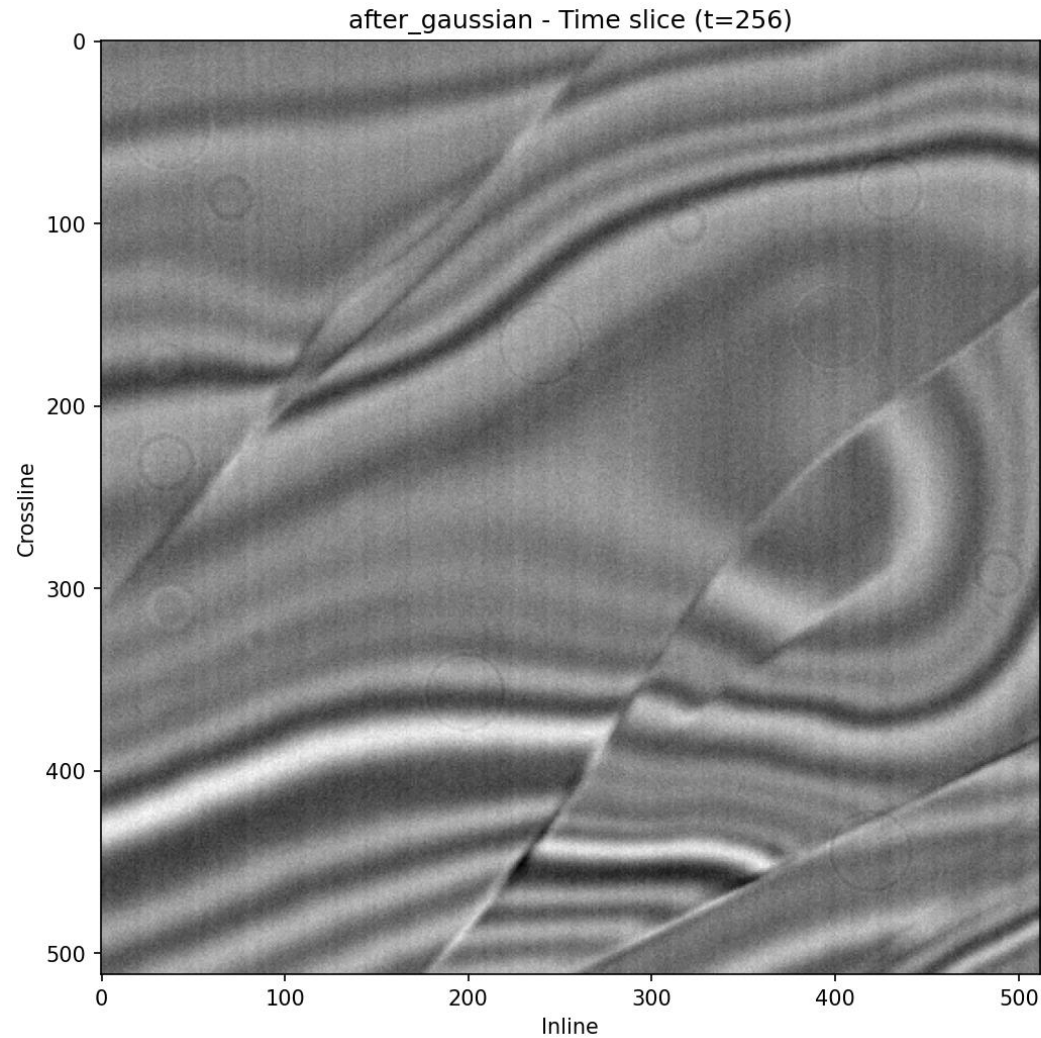
Footprints + Gaussian Noise



Artifacts + Footprints + Gaussian Noise

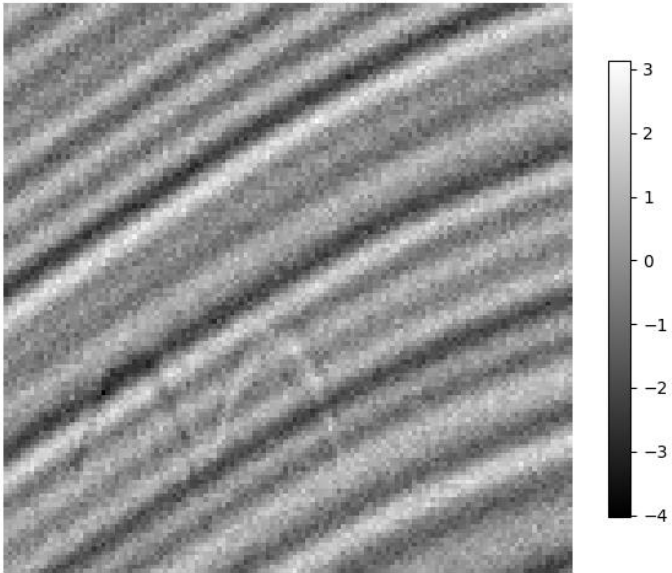


Artifacts + Footprints + Gaussian Noise

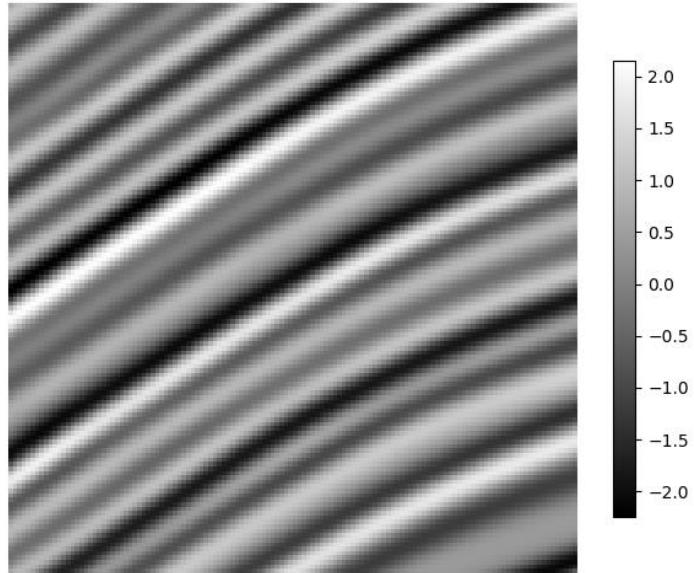


Denoised

Low Quality (LQ)



Ground Truth (GT)



Model Output

